Program - 9: Perform Testing Activities within an Agile Project

Aim: To study and perform software testing activities in an Agile project environment by developing a small feature, writing test cases, executing tests, identifying defects, and demonstrating how testing is integrated into each iteration/sprint.

Objectives

1. To understand how testing is embedded in Agile sprints.
2. To plan and execute test cases concurrently with development.
3. To practice writing user stories and their acceptance criteria.
4. To run manual/automated tests as part of Agile iteration.

Problem Statement

In traditional software development, testing occurs at the end of the lifecycle, leading to late detection of defects and increased costs. Agile requires continuous testing within each sprint to ensure working, high-quality increments. This experiment focuses on implementing testing activities (planning, execution, automation) within an Agile iteration.

Description

In Agile projects, testing is not a separate phase but an ongoing activity performed throughout the sprint. Testers, developers, and product owners collaborate to define acceptance criteria for user stories.

Testing activities include:

- Writing test cases alongside user stories.
- Performing unit testing, integration testing, and acceptance testing within the sprint.
- Using automation tools (like Pytest, Selenium, JUnit) to ensure quick feedback.
- Continuous integration ensures tests run automatically on each commit.

This experiment demonstrates how to take a small user story, write its acceptance criteria, create test cases, execute them (manually or automatically), and record results within a sprint.

Procedure

1. Select a User Story

In Agile, testing is based on user stories. Choose a small functionality:

- **User Story:** *"As a user, I want to calculate the sum of two numbers so that I can get the result quickly."*

Define Acceptance Criteria

Document clear conditions for the story to be considered "Done":

- The system should accept two numbers.
- It should return their correct sum.
- It should work with positive numbers.
- It should work with negative numbers.
- It should work when one number is zero.

2. Identify Test Cases

Test Case ID	Input	Expected Output
TC01	2, 3	5
TC02	-2, -3	-5
TC03	5, -2	3
TC04	0, 5	5
TC05	0, 0	0

3. Write Test Cases

Prepare test scenarios and cases based on acceptance criteria. Write the automated test cases.
Create another file named test_calc.py:

```
import unittest
from calculator import add

class TestCalculator(unittest.TestCase):
    def test_positive_numbers(self):
        self.assertEqual(add(2, 3), 5)

    def test_negative_numbers(self):
        self.assertEqual(add(-2, -3), -5)

    def test_positive_negative(self):
        self.assertEqual(add(5, -2), 3)

    def test_zero(self):
        self.assertEqual(add(0, 5), 5)

    def test_both_zero(self):
        self.assertEqual(add(0, 0), 0)

if __name__ == '__main__':
    unittest.main()
```

4. Develop the Feature

Developer implements the story during the sprint. Create a Python file named calc.py:

```
def add(a, b):
    return a + b
```

5. Perform Unit Tests

Run unit tests immediately after coding.

6. Analyze Test Results

The test framework checks whether:

\text{Actual Output} = \text{Expected Output}

- **If both are equal:** tests passed
- **If both are different:** test failed

7. Perform Integration Tests

Verify the new functionality integrates with existing modules.

8. Perform Acceptance Tests

Validate story against acceptance criteria with the Product Owner.

9. Perform & Automate Tests (Optional)

Use an automation tool (e.g., Selenium / Pytest) for repetitive checks.

10. Continuous Testing

Testing is performed continuously during development.

11. Log and Track Defects

Report issues in Agile board (Jira / Trello).

12. Review during Sprint Review

Demonstrate the working story with test results.

Output

```
. . . . .
Ran 5 tests in 0.001s

OK
Py: OK (1.06 seconds)
```

```
Congratulations :) (1.11 seconds)

> Python -m pytest
= = = = = = = = = = test session starts = = = = = = = = = =
Platform win32 -- Python 3.14.6, pytest - 9.1.1
Pluggy - 1.6.0
Collected 5 items

test_calc.py . . . . .                                  [100%]

= = = = = = = = = = 5 passed in 0.02s = = = = = = = = = =
```

Conclusion

Testing activities were successfully performed within the Agile project during the sprint. This demonstrated that Agile testing is continuous, collaborative, and aligned with user stories, ensuring high quality and quick feedback for each increment of software.